Advanced Computer-Aided VLSI System Design

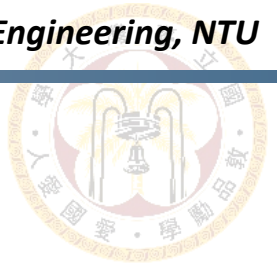
Chapter 9. Physical Sign-Off

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Graduate Institute of Electronics Engineering, National Taiwan University

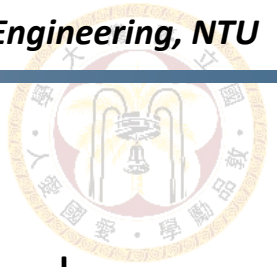


NTU GIEE



Outline

- Process Overview
- Design Rule Check
- Layout Versus Schematic
- Dummy Fill
- Design for Manufacturing and Physical Cell

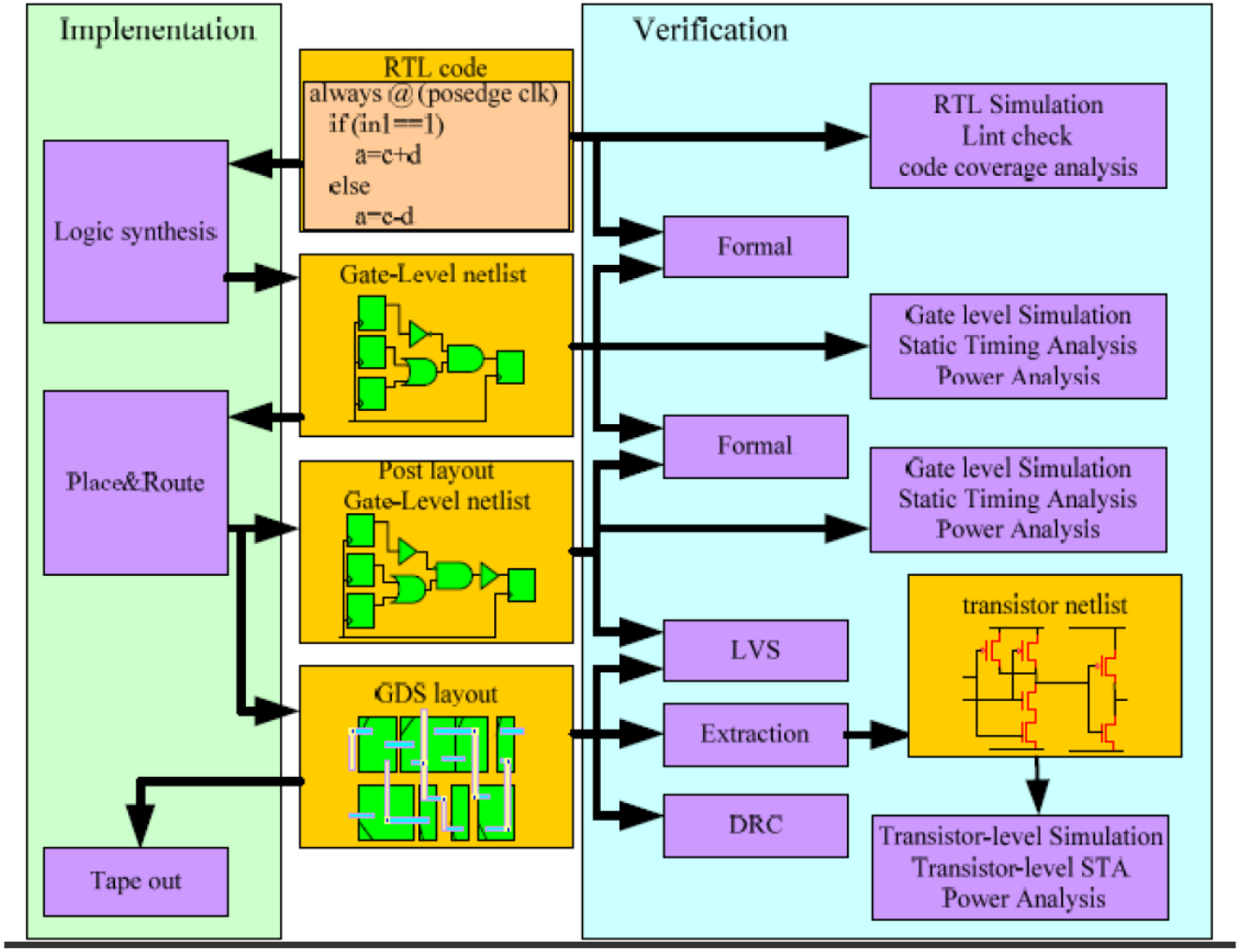


Important

The design rule is highly related to each factory's process and technology, and it should be strictly controlled.

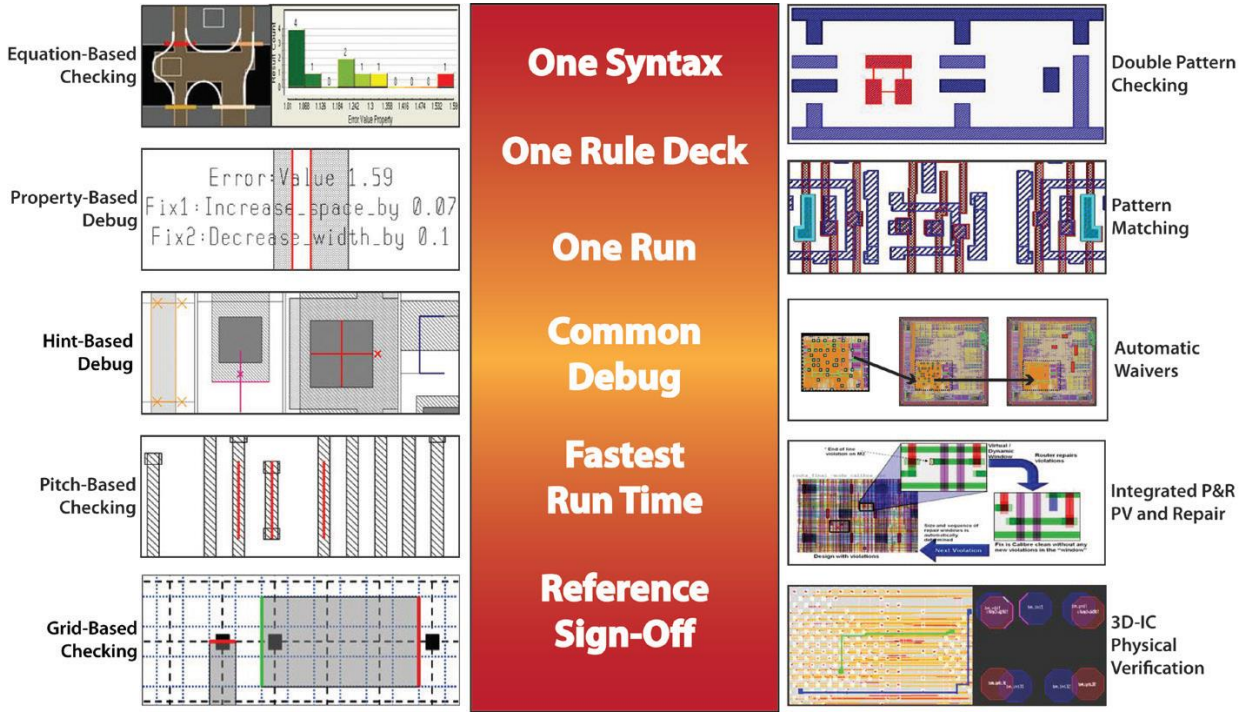
Please DO NOT release any information about DRC or the physical design to others

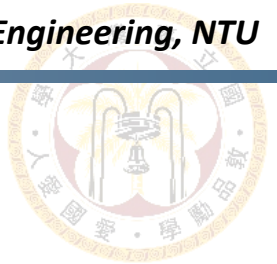
Cell-Based Design Flow



Physical Verification (PV)

- Functional design verification is only for designer
- Physical design must meet process rules for **manufacture reliability**
- Even commercial placement & route tools can introduce error
- The performance of design after layout needs to be verified



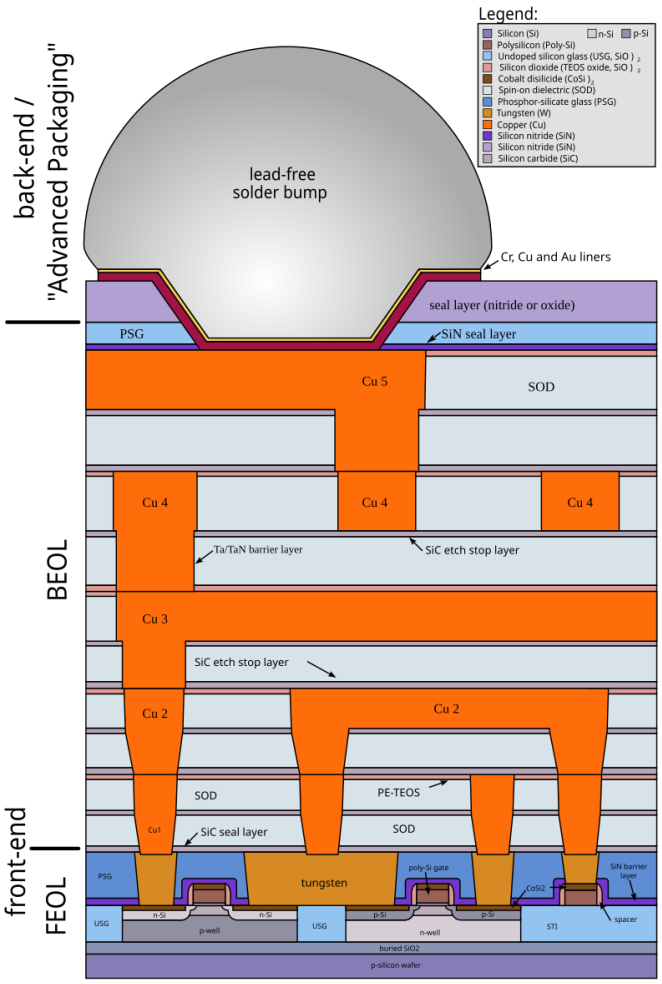


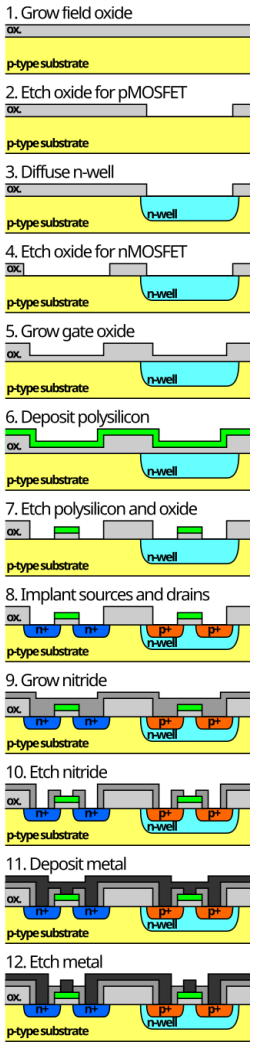
Outline

- **Process Overview**
- Design Rule Check
- Layout Versus Schematic
- Dummy Fill
- Design for Manufacturing and Physical Cell

Integrated Circuit Cross Section

Planer IC





FinFET Structure

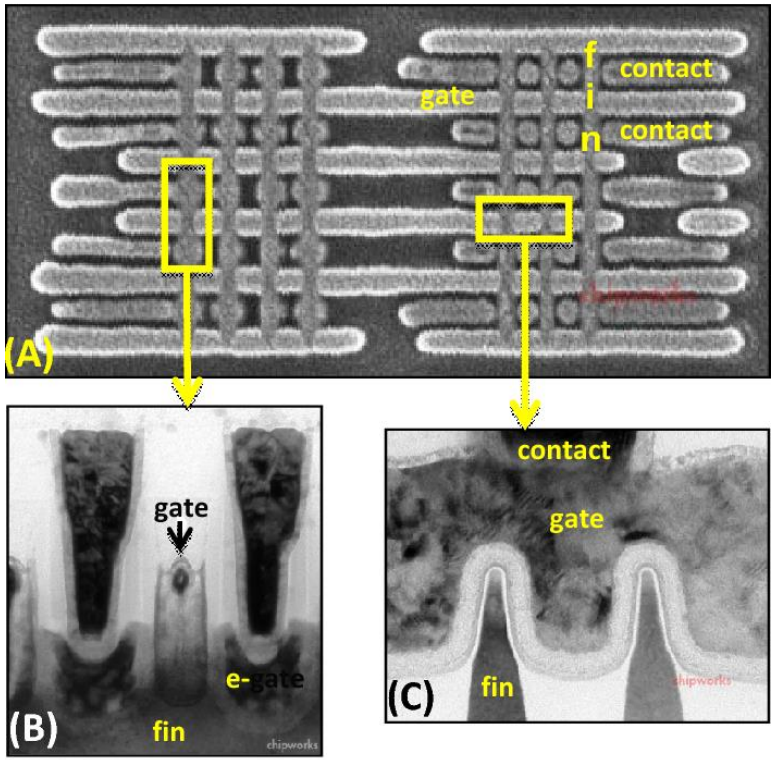
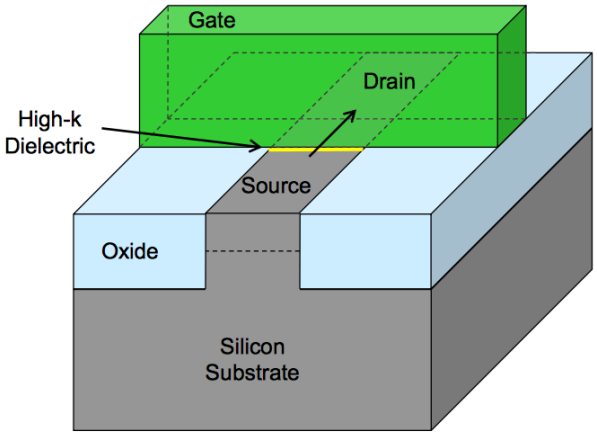
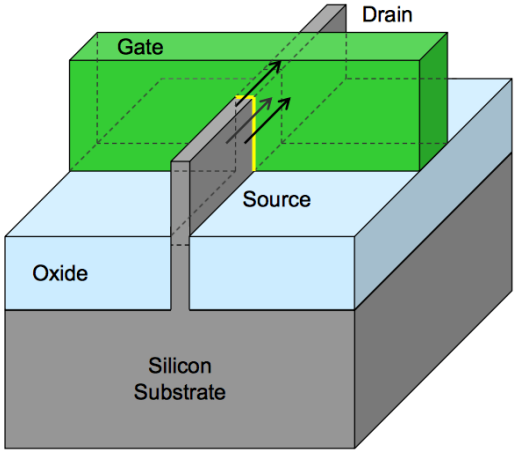
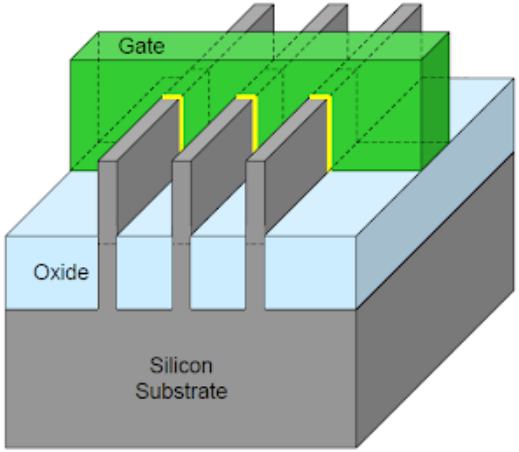


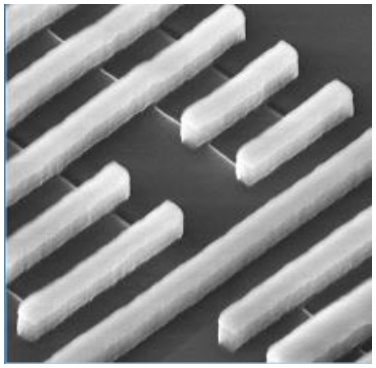
Fig. 7 (A) SEM plan-view image of Intel 22-nm transistors; (B) TEM cross-

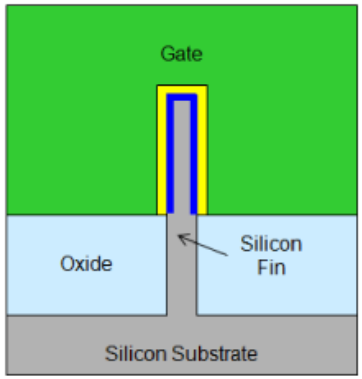
FinFET Architecture

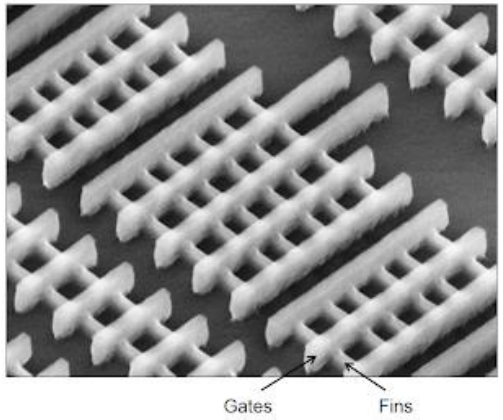






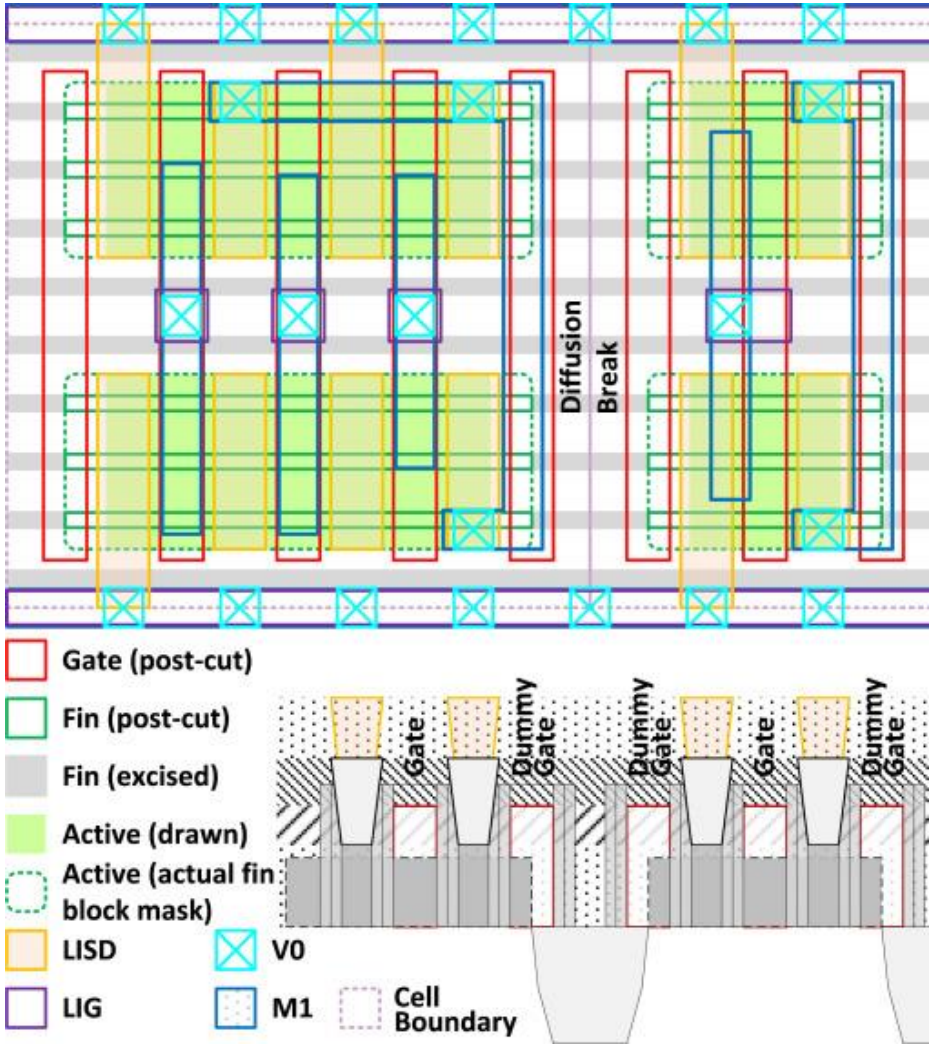






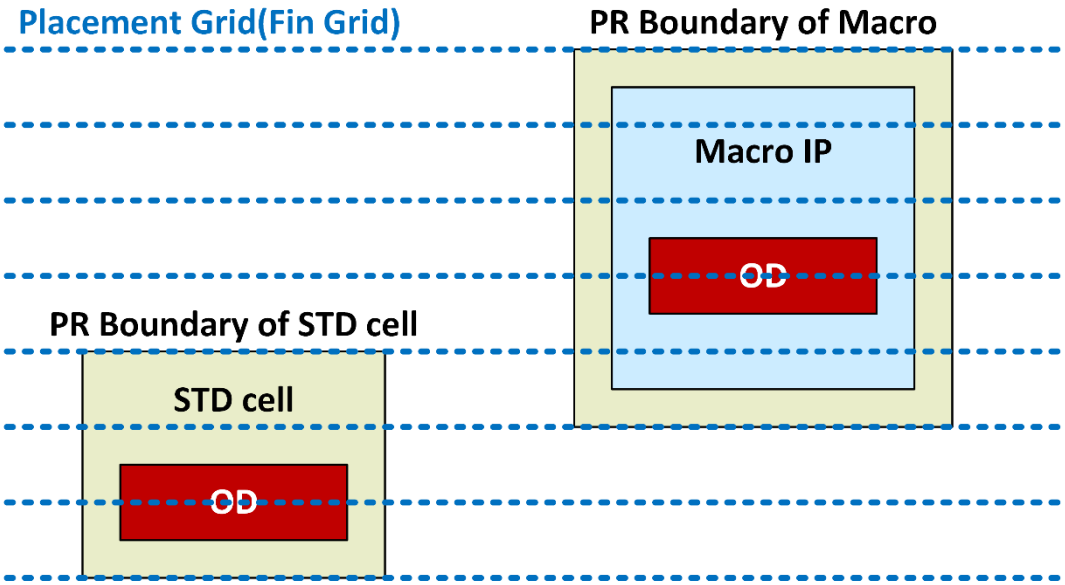
Planar Transistor **FinFET Transistor** **Multi-Fins Transistor**

FinFET Layout View



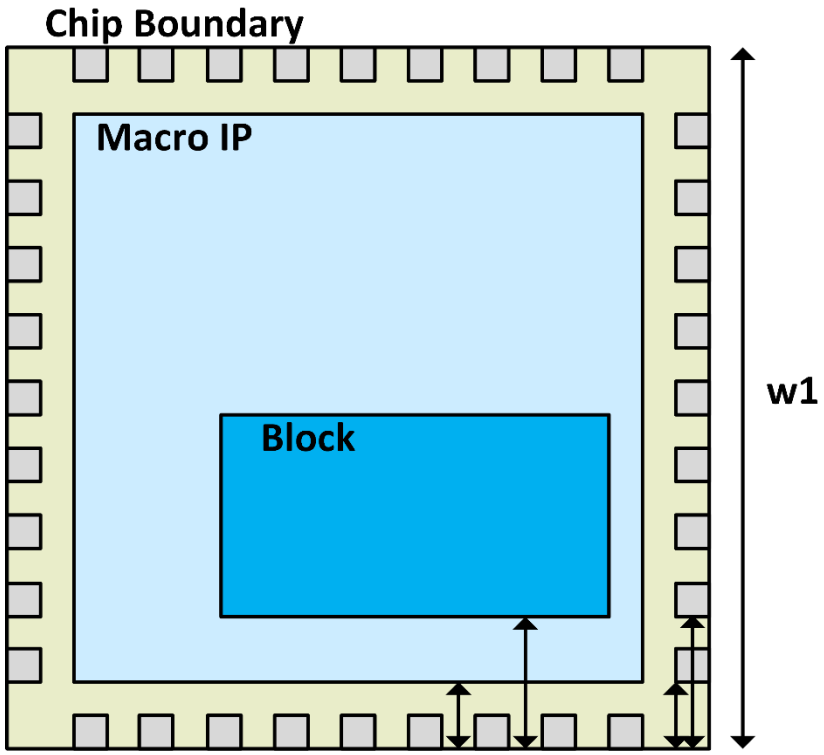
Placement Grid

- All OD (Gate Oxide and Diffusion) regions in the std cell and macro should be placed on the FinFET grid
- Methodology
  - Create the cell boundary while creating a macro
  - Cell/macro placement follows a universal placement grid



Placement Grid

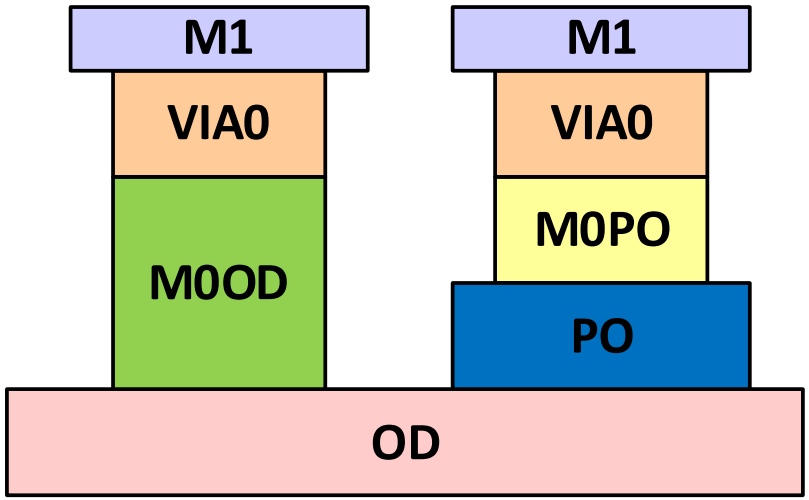
- Design partitions need to cover integration conditions
- The chip boundary need to be the even integer of the Fin pitch
  - W1 = 0.288 + 0.096 * n



Front-End/Middle-End of Line

- Front-end of line (FEOL) includes all steps of formatting transistors
- Middle-end of line (MEOL) dealing with the first-level interconnects and contact/via formation
- MEOL layers not only serve as the bridge between FEOL and BEOL, but also as the local connection of the circuit

ADFP/N16 (FinFET)



M1: metal 1

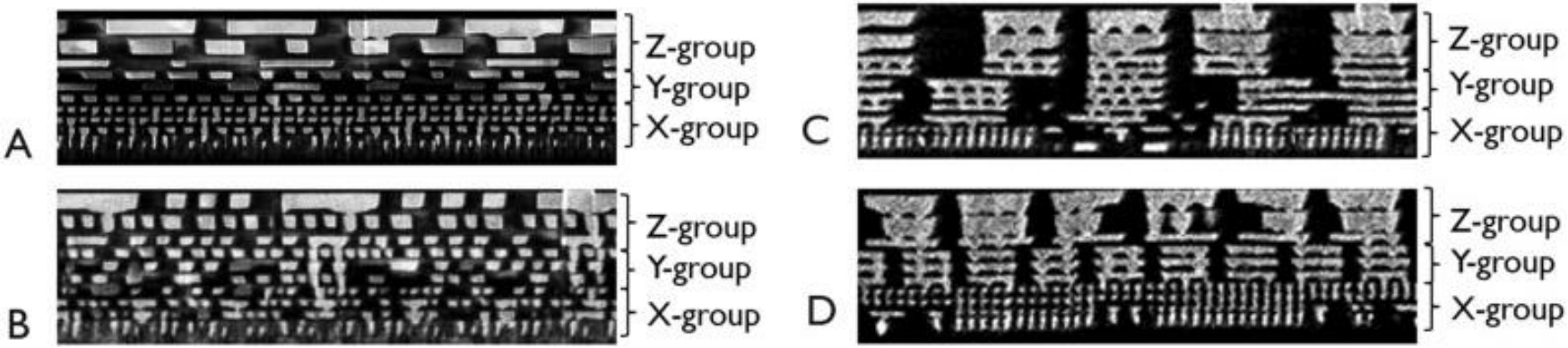
M0: metal 0

OD: Gate Oxide and Diffusion

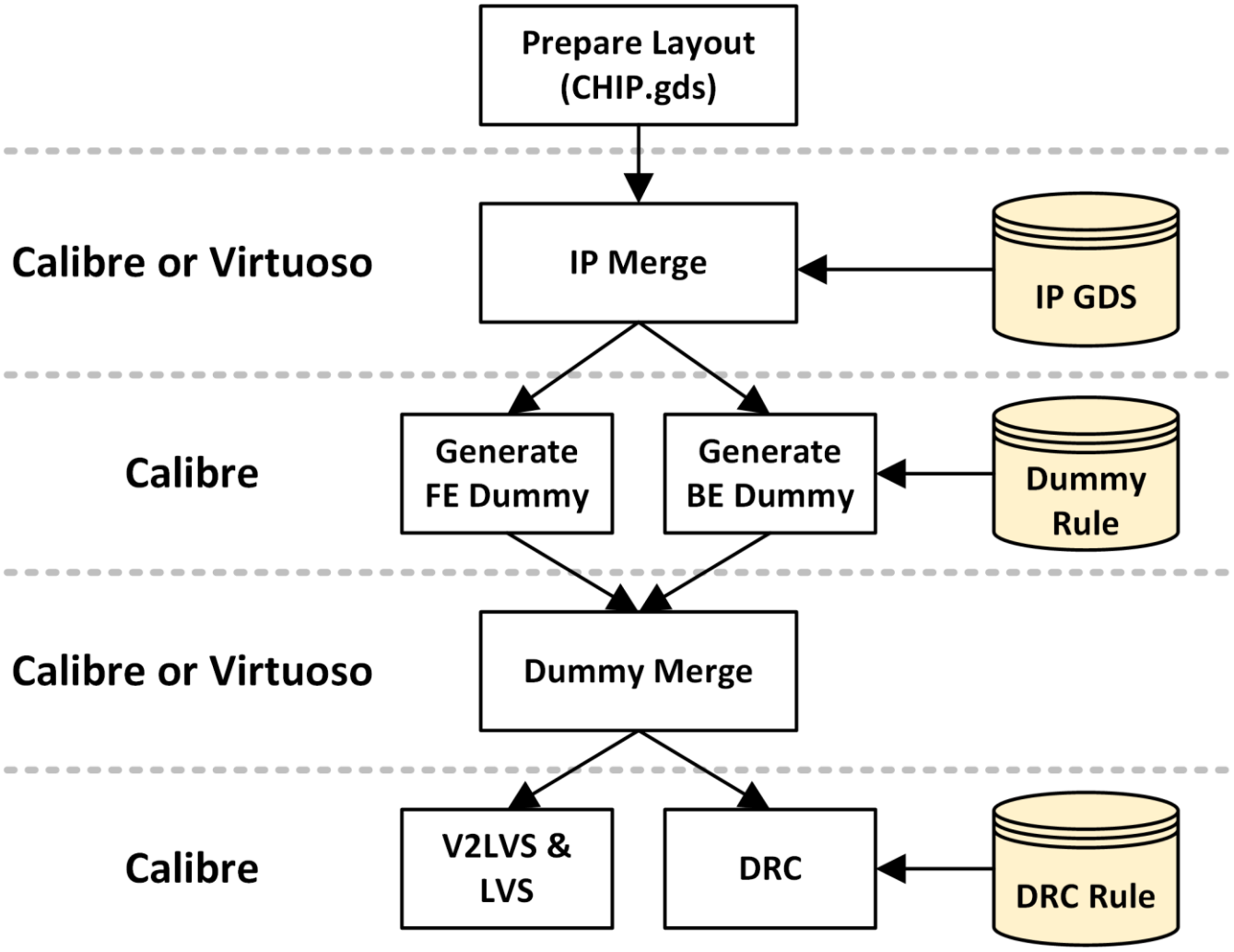
PO: Poly-Silicon

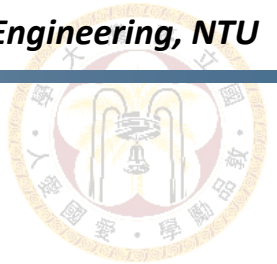
Back End of Line

- There are various metal types in a process
  - Mx, My, Mz…
- The layers in the same metal type are produced by the same progress and have similar property
- We can recognize the metal stack up from the document
  - ADFP: 1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R



Physical Verification Flow





Outline

- Process Overview
- **Design Rule Check**
- Layout Versus Schematic
- Dummy Fill
- Design for Manufacturing and Physical Cell

Design Rule Check (DRC)

- DRC is an automated verification process to ensure an IC layout complies with all the manufacturing rules
- Why run DRC after layout
  - **Manufacturability:** Ensures the design can be fabricated reliably
  - **Reliability:** Prevents shorts, opens, and long-term failures
  - **Higher Yield:** Reduces defect rates, boosts good dies per wafer
- The DRC check should be run during and after ARP

APR Design Rule Check

- The design rule check during APR
- Tools: **Innovus**, ICC2
- The design rule is defined in the technology file (.lef)

```
********** Technology File **********
LAYER ${Layer Name}
    TYPE IMPLANT;
    WIDTH ${Min Width}
    SPACING ${Min Spacing}
    PROPERTY LEF58_AREA”
    AREA ${Min Area};”;
END ${Layer Name}
```

Sign-Off Design Rule Check

- The DRC information of the sign-off check is from the DRC table
  - DRC.encrypt
- The information of the design rule is included in the DRM
  - The design rule
  - The information about the process
  - The layer mapping of the process
  - The stack up options
- Tools: **Calibre**

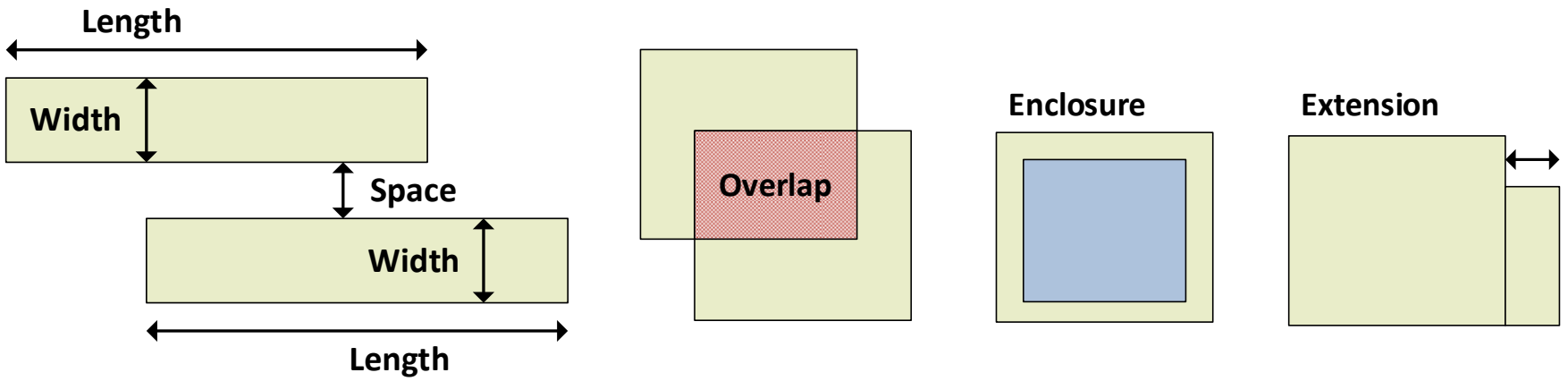
Design Rule Manual

- DRC rule: the basic unit is um; unit of area is um^2
- Layout/DRC dimension must be an integer multiple of 0.001 um
- Unless otherwise specified, all rules specify minimum dimensions
- Design rule requires exact dimension ("=") is not to be relaxed
- Bracket means:
  - () means explanations
  - [] means under special/certain conditions
  - {} means logic operations for DRC

Rule	Description	Label	Op.	Rule
PO.S.13	TrGATE to 1st PO space rule (Lg=0.0xxum):{{TrGATE SIZING 0.0xxum} AND PO} space to CrtPO of core device [channel length = 0.0xxum]	S13	=	0.0xx

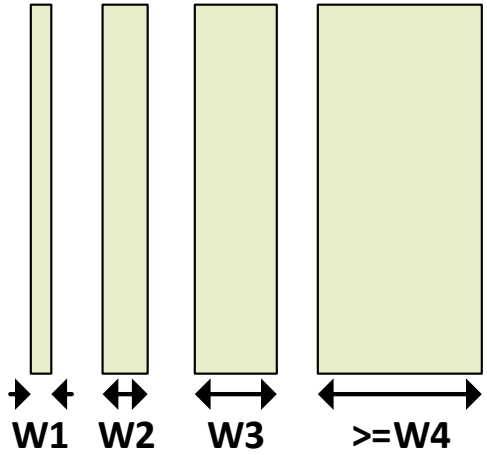
Typical DRC

- 3 common rules: Width (**W**), Length (**L**), Space (**S**)
- Example: Mx.W.1
  - The **first** rule of the **width** of **x type of metal**
- Other rules: Overlap (**O**), Enclosure (**EN**), Extension (**EX**), Area (**A**), Enclosed Area (**A**), Density (**DN**), Parallel run length
- Other types of DRC will be represented with the symbol **R**

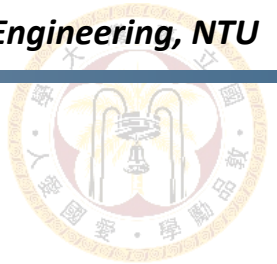


Discrete Metal/Poly Width

- Width of metals/poly is discrete under a certain value
- Wires on preferred/non-preferred direction with different sets of routing pitch, metal spacing, EOL spacing rule



Valid width (um)	Discrete	Continuous when
Preferred direction	[1.xx], [2.xx], [3.xx]	>=[6.xx]
Non-preferred direction	[3.xx], [5.xx], [8.xx]	>=[10.xx]

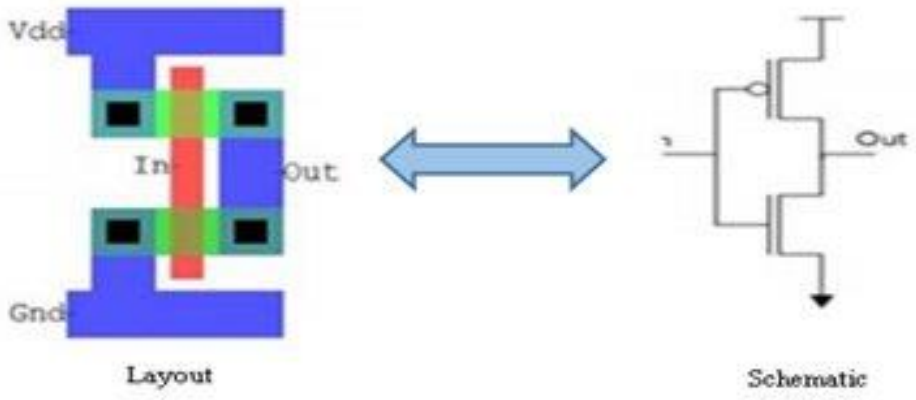


Outline

- Process Overview
- Design Rule Check
- **Layout Versus Schematic**
- Dummy Fill
- Design for Manufacturing and Physical Cell

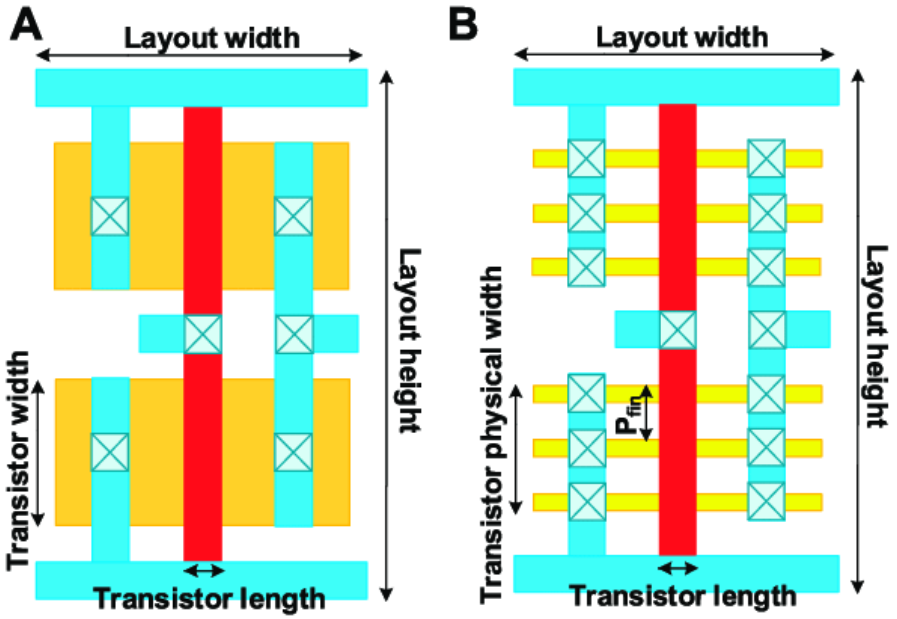
Layout Versus Schematic

- Layout Versus Schematic (LVS) check whether the layout is the same as the schematic (Verilog netlist in cell-based design)
- Common check errors:
  - Net open/short
  - Net missing/redundant
  - Object missing/redundant
- Aspects: transistor-level LVS, block-level LVS



Transistor-Level LVS

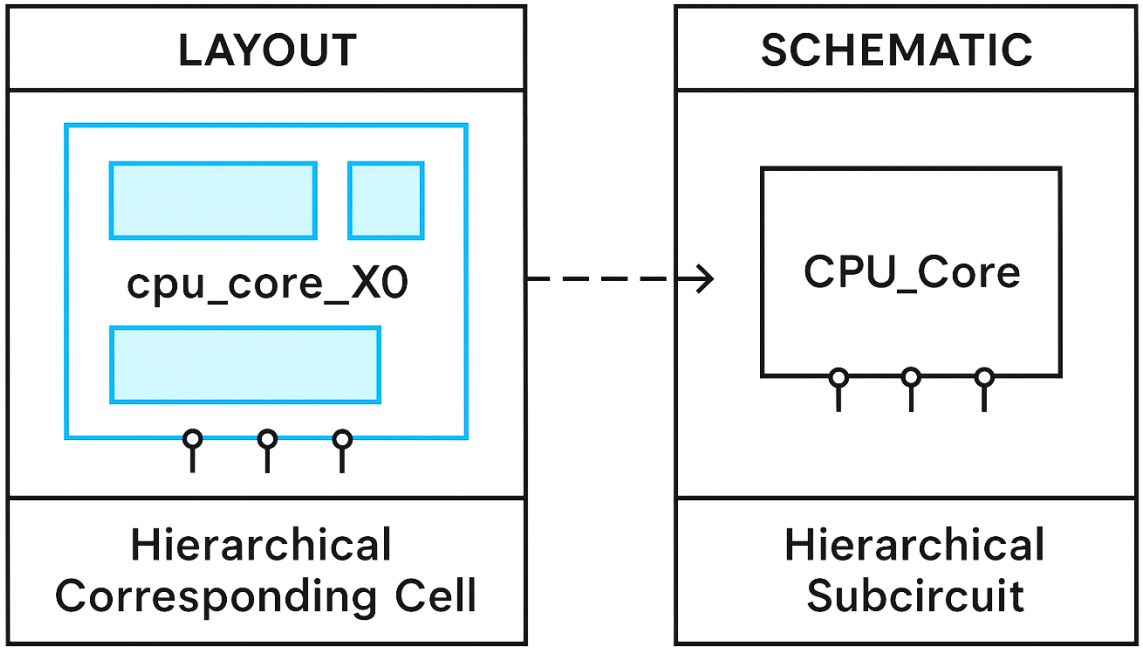
- Check the connectivity of essential elements
- The elements in the layout are automatically recognized by tools
  - No need to annotate the position of the elements
- Transistor-level LVS mainly applies to **full-custom designs**



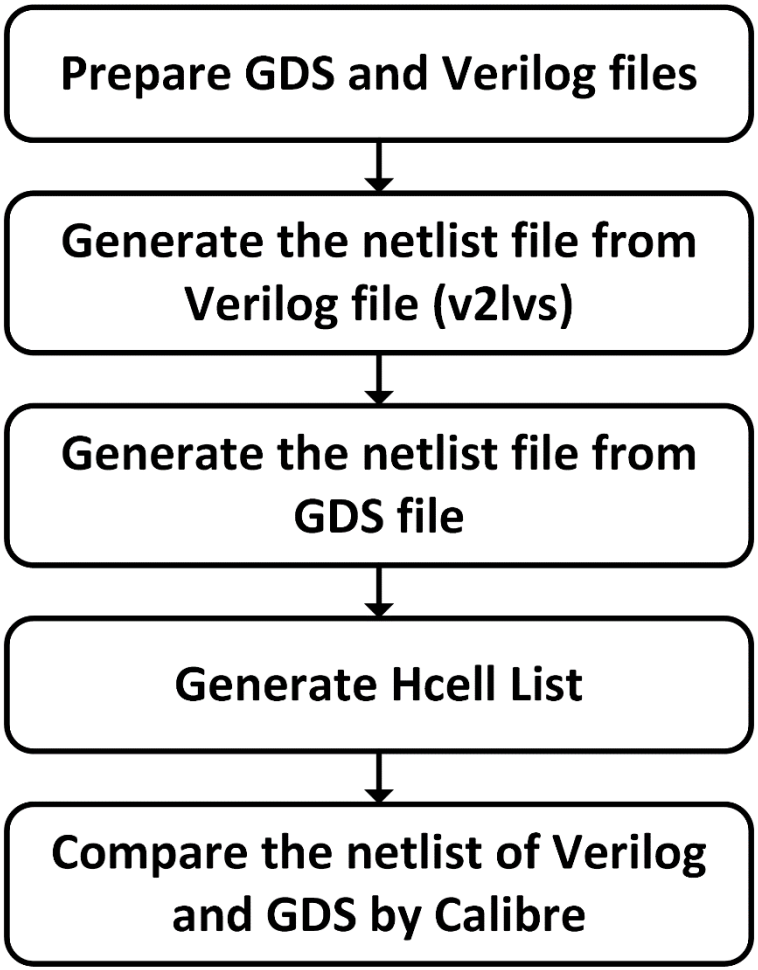

```
********** spice netlist **********
.GLOBAL VDD GND
.SUBCKT inv GND VDD a y
M1 y a GND GND nch nfin=3 L=0.016u
M2 y a VDD VDD pch nfin=3 L=0.016u
.ENDS
```

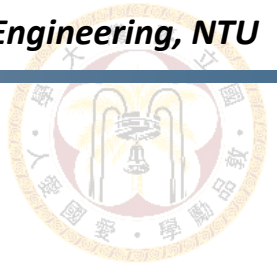
Block-Level LVS

- Check the connectivity of standard cells and macros
- Applying a hierarchical layout instead of the detailed layout
- Reduce the computation requirement of the LVS check
- Block-level LVS mainly applies to **cell-based designs** or IPs



The Flow of Block-Level LVS





Layout and Netlist Preparation

- The netlist should contain all **logical** and **physical** cells
  - Modify the write_netlist script of Innovus

```
# ********** write_netlist.tcl File **********
saveNetlist -includePowerGround -includePhysicalCell "[all
physical cell list]" -excludeLeafCell output/CHIP_pg.v
```

- The GDS file should be merged with all standard cells and macros GDS
  - Merge GDS files by Innovus

```
# ********** write_stream.tcl File **********
streamOut output/CHIP.gds -mapFile [map_file] -merge [all gds
files] -uniquifyCellNames -unit 1000 -mode all
```

  - Merge GDS files by Calibre
  - Merge GDS files by Virtuoso

Hierarchical Cell List

- Hierarchical cell (Hcell) refers to sub-blocks of the netlist and the leaf cells of the layout
- The Hcell list can be specified in Calibre script or in an Hcell document

```
# ********** Hcell list in Calibre script **********
LVS BOX [layout cell name1] [netlist cell name1]
LVS BOX [layout cell name2] [netlist cell name2]
LVS BOX [layout cell name3] [netlist cell name3]
```

```
# ********** Hcell list in Hcell document **********
[layout cell name1] [netlist cell name1]
[layout cell name2] [netlist cell name2]
[layout cell name3] [netlist cell name3]

# ********** In Terminal **********
> calibre -hcell [Hcell list] ...
```

Generate the SPICE Netlists

- Generate the netlist from the Verilog

```
> v2lvs –v [Verilog design] –l [standard cell Verilog file
(with power/ground net)] –s [spice list] -o CHIP_subckt.spi
```

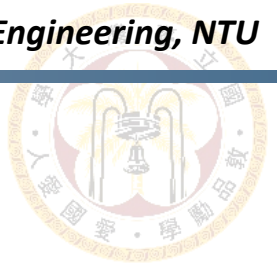
```
********** spice netlist **********
.SUBCKT CHIP [port list]
X1 [net list] SUB_MODULE_1.ENDS
```

- Generate the netlist CHIP.layspi from the GDS

```
> calibre –hcell [hcell document] -64 –hier –hyper –spice
./output/CHIP.layspi ./scr/runset.cmd
```

```
********** spice netlist **********
.SUBCKT CHIP [port list]
X1 [net list] SUB_MODULE_1 $T=[x pos.] [y pos.] [Mirror/Flip]
[rotation] $X=[x position] $Y=[y position]
.ENDS
```

- Then, you can run the LVS check 😎

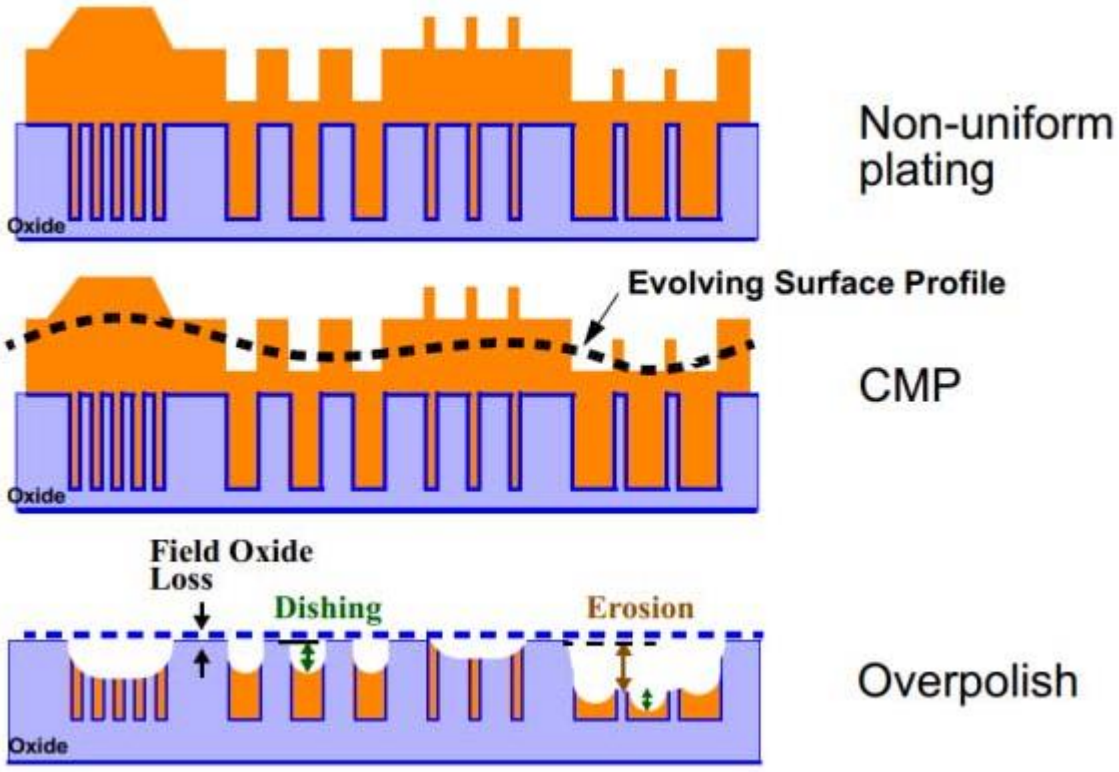


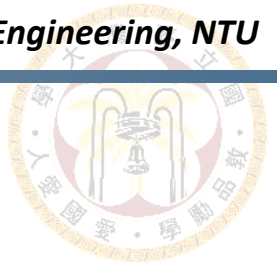
Outline

- Process Overview
- Design Rule Check
- Layout Versus Schematic
- **Dummy Fill**
- Design for Manufacturing and Physical Cell

Over Polish

- Excess CMP time removes metal below the target level
- Dishing: wide metal features polish faster than field oxide
- Erosion: dense arrays lose oxide/metal height and planarity



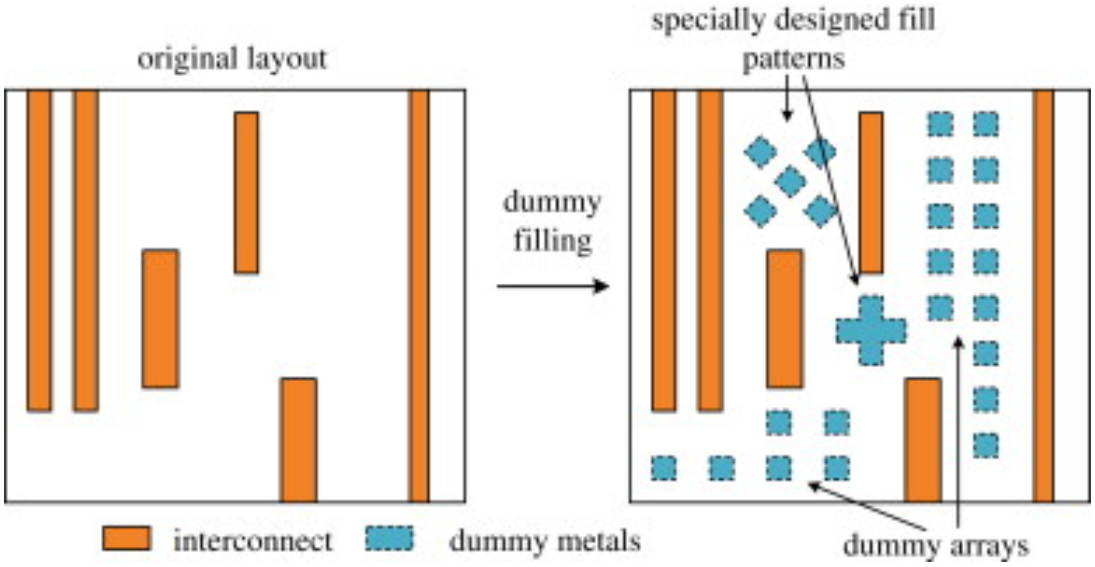


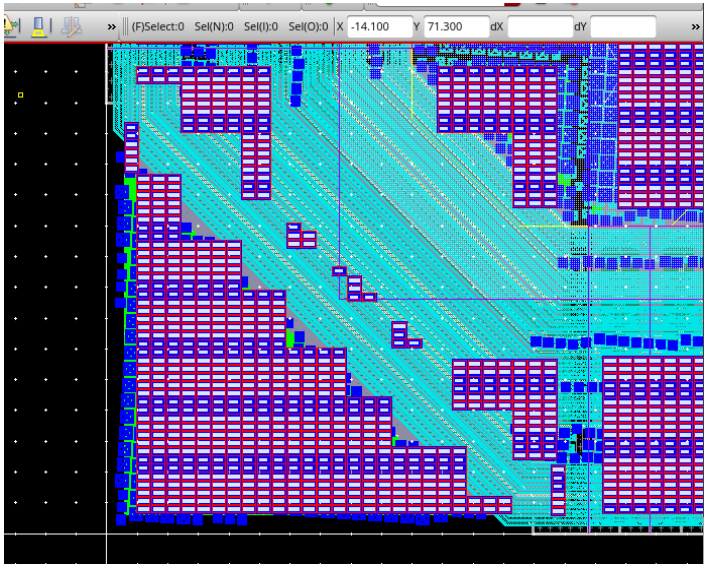
Dummy Fill

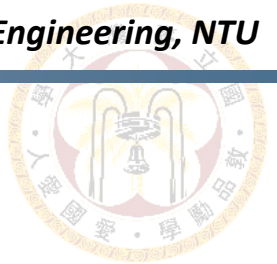
- Meet the metal density requirement
- Add fill according to the density gradient
- Addressing rapid thermal annealing (RTA)
- Addressing manufacturing stress
- Addressing etch density
- Tool: **Calibre nmDRC**

FEOL/BEOL Dummy Fill

- Calibre will automatically generate the dummy fill according to the density requirement of the process
- The dummies are filled in the dummy layer







Outline

- Process Overview
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- **Design for Manufacturing and Physical Cell**

Physical Only Cells

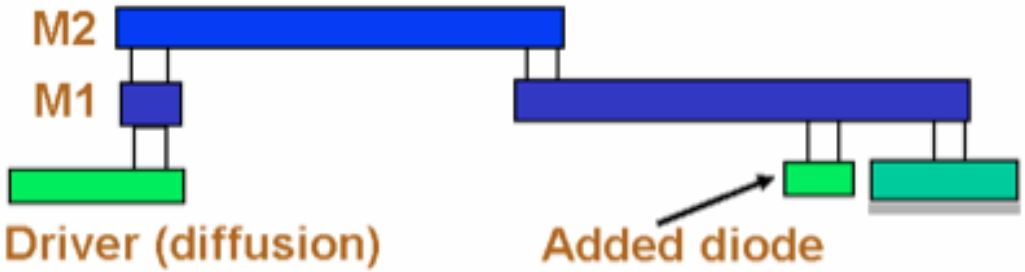
- The physical cell won't effect the netlist
- The physical cell present in the design netlist
- Physical cells for **preventing damage while manufacturing**
  - Antenna, End-cap
- Physical cells for **ECO**
  - Spare cells
- Physical cells to make sure **standard cells work normally**
  - Well-tap, Core-filler, Decap
- Physical cells for other purpose
  - Tag cell, Tie cell

Antenna Rule

- In a chip manufacturing process, metal is initially deposited so it covers the entire chip
- The unneeded portions of the metal are removed by etching
- The exposed metal collect charge and form voltage potential
- If the voltage potential across the gate oxide becomes large enough, the current can damage the gate oxide
- Problem repair
  - Add jumper (change metal layer)
  - Add antenna cell (diode)
  - Add buffer

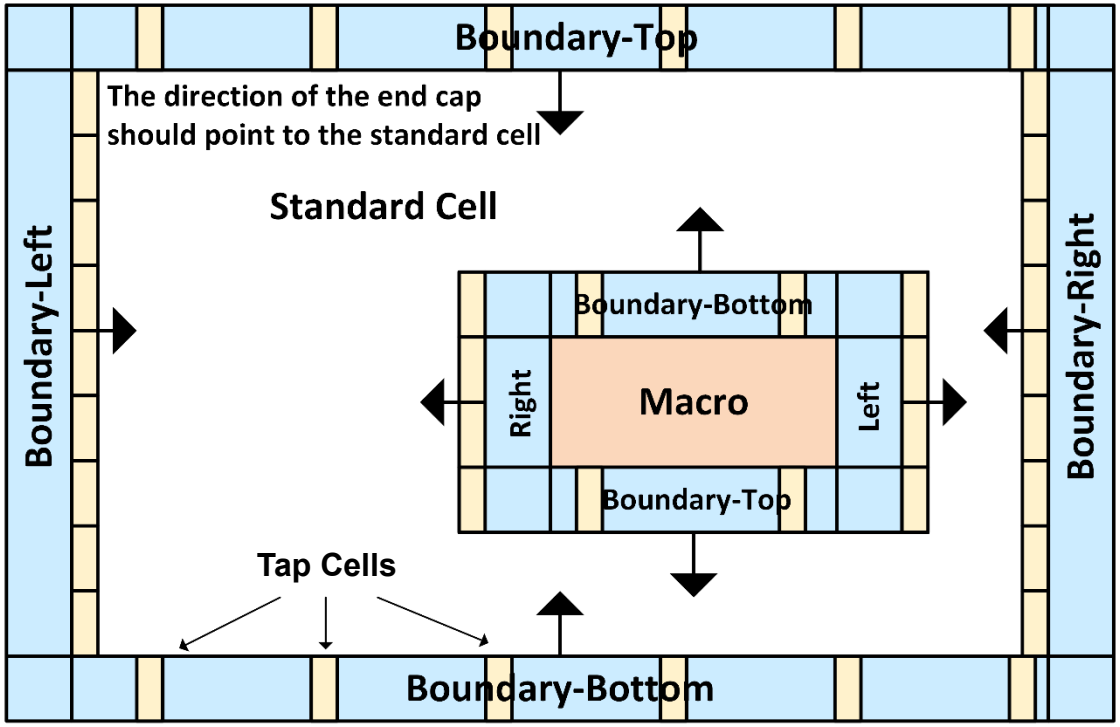
Antenna Cell

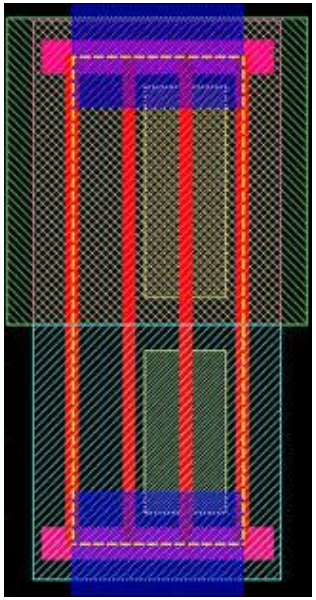
- The antenna cell can discharge the static electron and the voltage potential across the gate oxide
- When the accumulated charge is large enough, the diode will be short



End Cap

- End cap cell, also known as a boundary cell
- Placed at the edges of a chip and macros to protect standard cells
- In ADFP, the boundary cell should surround the standard cells



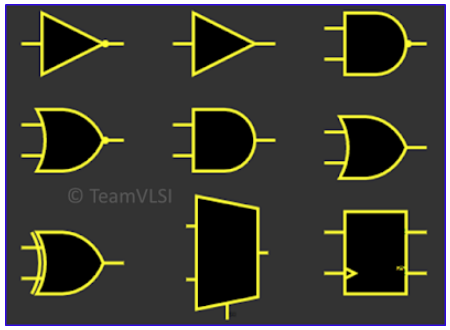


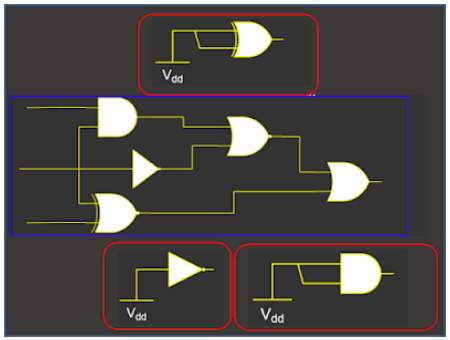
End Cap layout of planer technology

source: team VLSI

Spare Cells

- Do not perform any logical operation in the design
- Change the function easily by changing the metal (Metal ECO)
- Standard cells: inverter, buffer, nand, nor, …





- Floating inputs introduce additional switching and dynamic power
- The input of the spare cells should be connected to the supple/ground



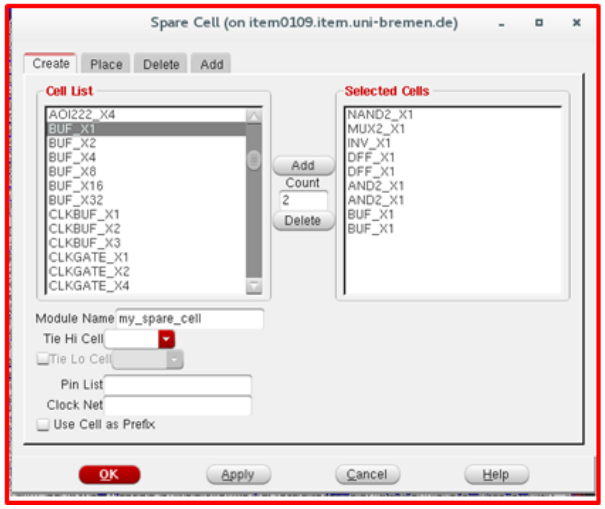
Spare Cells

- Add before the placement
- Usually use the cells with minimum sizes

Command

```
setPlaceMode -moduleAwareSpare true
createSpareModule -moduleName spareModule -cell {INVX2 INVX4 BUFX2 NAND2X1 DFFPOSX1 ... } -tie TIE0
placeSpareModule -moduleName spareModule -channel -minWidth 5 \
-maxWidth 10 -minLength 11 -stepx 10 -stepy 10 -util 0.7
```

GUI:

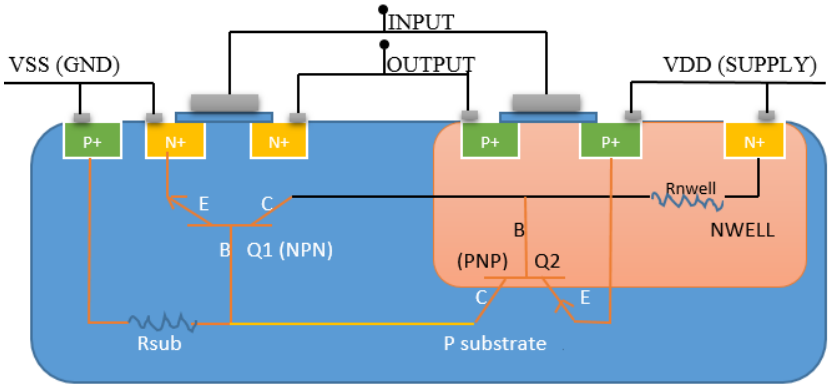


Spare Cells

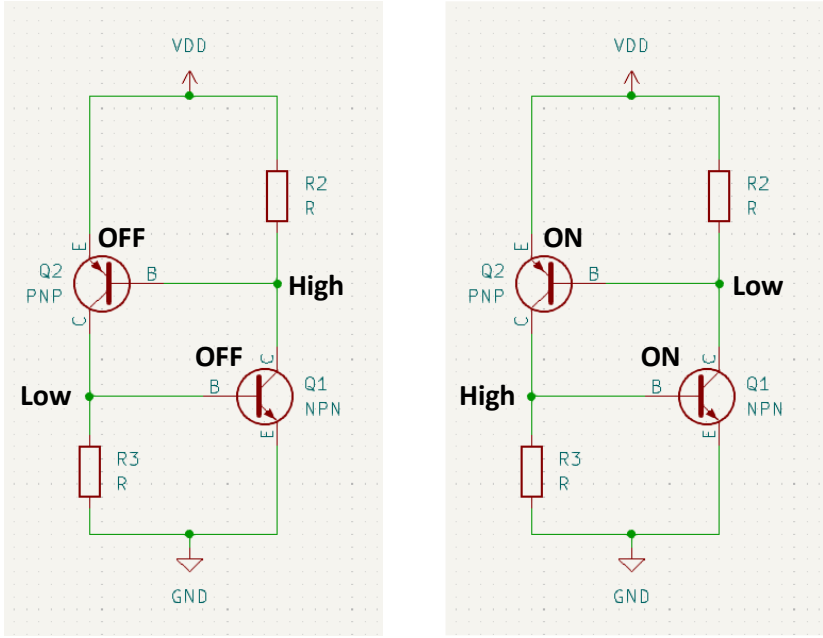
- **Pros:**
  - Reusability of FE masks and some metal masks
  - Small changes can be done very easily
  - Saving time and cost
- **Cons:**
  - Increase leakage power
  - Increase the area overhead
  - Occupy some routing resource

Latch-Up Phenomenon

- The P and N regions effectively form a back-to-back PNPN structure and become a latch if N-well/P-substrate resistances are great enough
- There are two stable situations for latch architecture



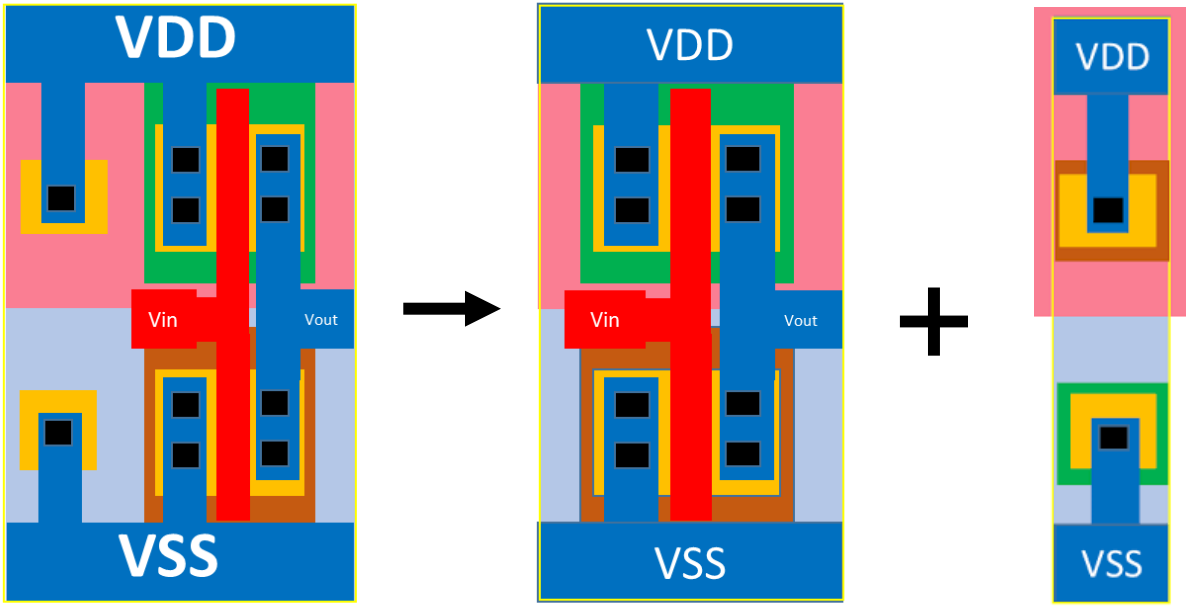
The PNPN structure in CMOS technology



The two stable situations: (left) works normally (right) consumes great current

Well Tap

- The resistance of the N-well/P-substrate should be less enough
- Tap-less STD cells have a smaller area but need well-tap cell
- TAP = Bulk = STRAP = Body
- The distance rule between the STD cells to well-tap is added to DRC



Traditional Standard Cell layout

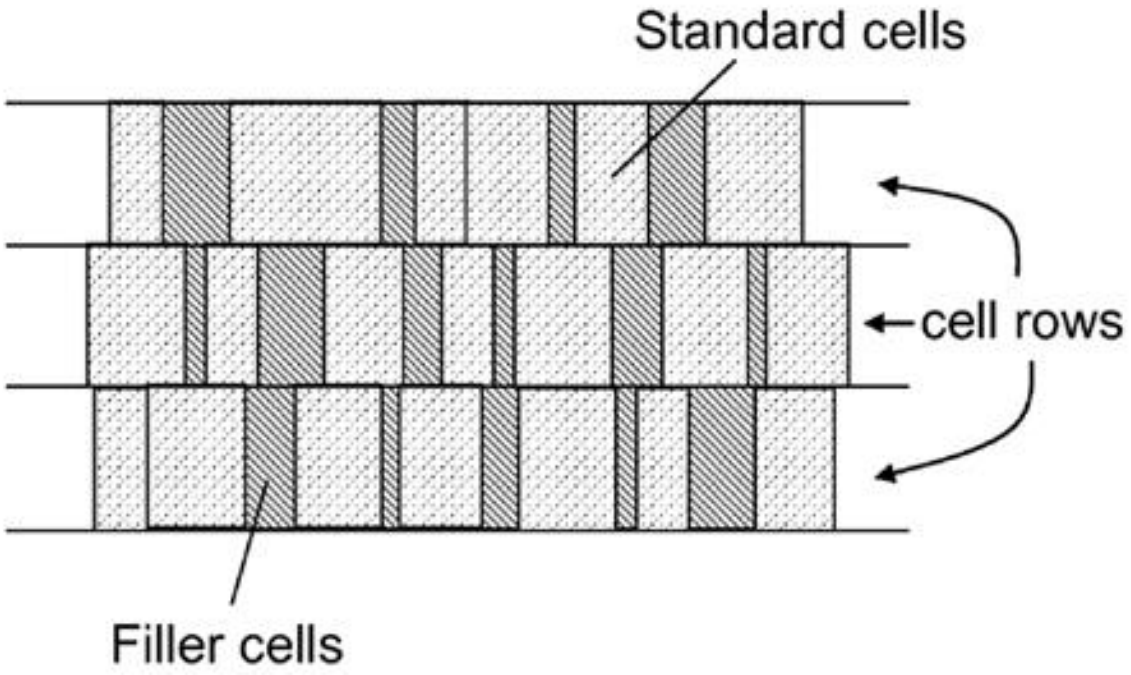
Tapless Standard Cell layout

Reference:

How to check whether the cell is tap-less

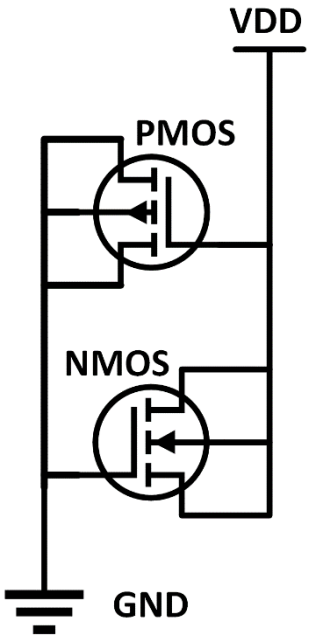
Core Filler

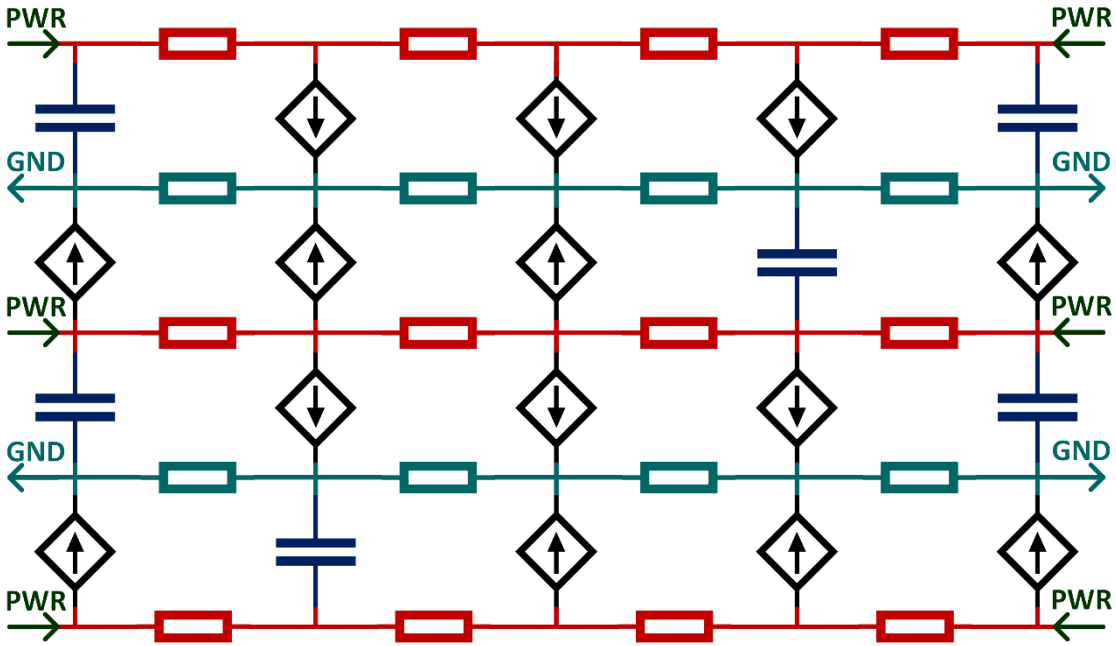
- Connect the N/P well of the standard cell
- Meet the requirement of the density of the FEOL and BEOL
- Connect the VDD/GND of the standard cell



DECAP

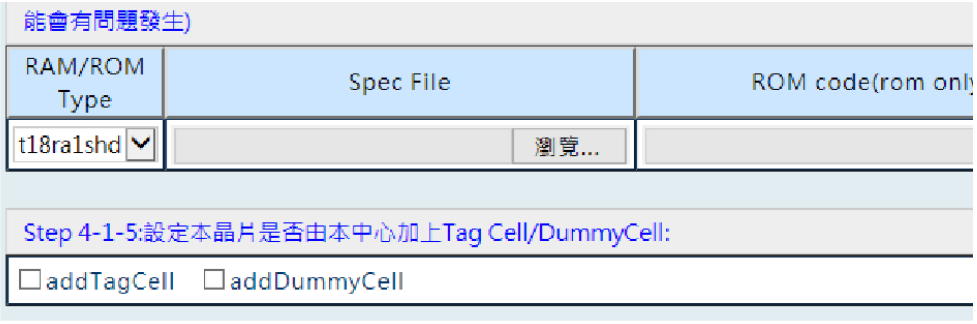
- Decoupling cell (DECAP cell) stabilizes the supply voltage
- It reduces the peak dynamic power and prevents the IR drop failure
- However, it increases the static current!!!
  - Only add the DECAP cell to the center/high-power area



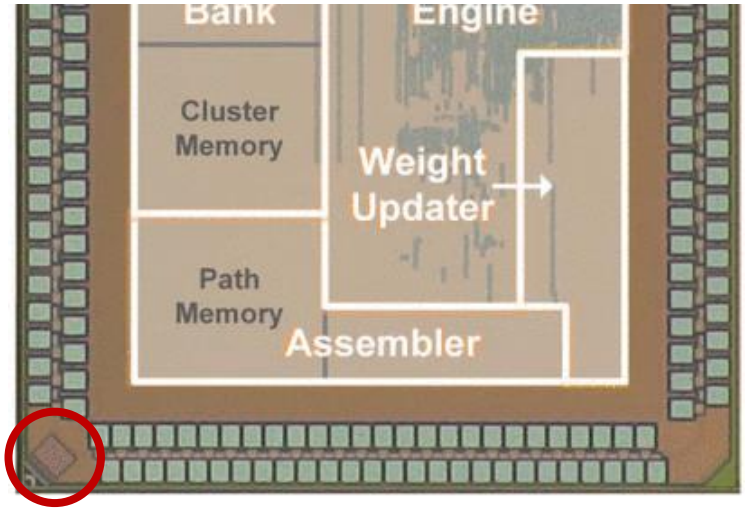


Tag Cell

- The tag cell is used to identify the direction of the chip for bonding and packaging
- Tag cell should be added/drawn to the **bottom-left corner** of the die
- The tag cell can be added manually by the layout tools (Laker/Virtuoso) or automatically by TSRI



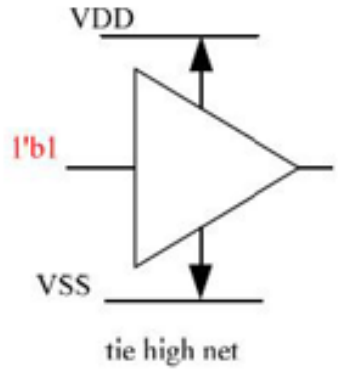
Automatically add tag cell by TSRI



Tag cell

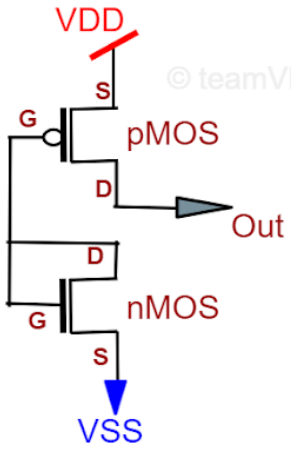
Tie Cell

- If the gate of the CMOS connects to the supply or ground directory, the gate may breakdown
- It will lead to permanent damage

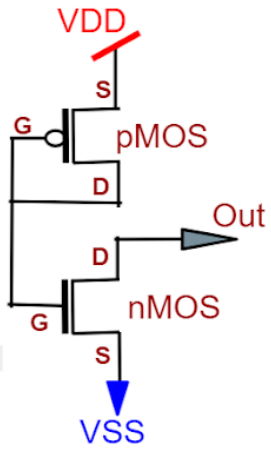


tie high net

```
INV inv1(.I(1'b1), .O(o));
```



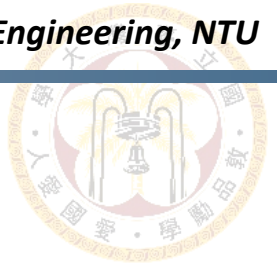
Tie-high Cell



Tie-low Cell

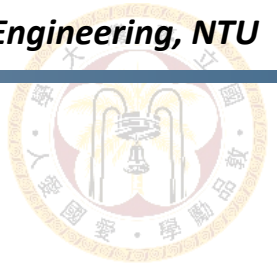
Rule Comparison

		Planar(0.13)	FinFET(N16)
Floorplan	Poly Direction	Bi-direction	**Uni-direction**
	Placement Grid	No Constraint	**Universal Grid**
	Logic Boundary	No Constraint	**Dedicated boundary cells**
	Well Tap	TAP embedded in every STD cell	Shared Well TAP to cover certain region
	Power Management	N/A	Common Well
Placement	Vt min area	No Constraint	Min-Vt Constrain
Routing	Metal Pitch	Same default pitch	Different Default Pitch
	Metal Width	No Constraint	Discrete width offering



Reference

- PHYSICAL ONLY CELLS - VLSI- Physical Design For Freshers
- Spare Cells in Physical Design - Team VLSI
- Making dummy fill smarter - Tech Design Forum Techniques



Appendix-Special Definition

- **PV:** Physical Verification
- **DRM:** Design Rule Manual
- **DRC:** Design Rule Check
- **PDK:** Process Design Kit
- **FEOL:** front end of line
- **BEOL:** back end of line

Appendix-Special Definition

- **FEOL**
  - **NW:** N-Well
  - **PW:** P-Well
  - **OD:** Oxide Definition (Active Region)
  - **DOD:** Dummy OD
  - **AP:** Aluminum Pad
  - **PO:** Poly-Silicon
  - **PP:** P+ implantation
  - **NP:** N+ implantation
  - **CO:** Contact from M1 to OD or PO
- **BEOL**
  - **Mx:** x type Metal
  - **VIAx:** x type VIA
  - **MOM:** Metal-Oxide-Metal (typically for a capacitor)
  - **CUP:** Circuit Under Pad